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2026–2027

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Wind and Solar Radiation Complementarity Analysis

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October 16, 2026

Abstract

This study examines the temporal and geographical patterns of complementarity between wind speed and solar radiation in Switzerland. Using hourly observations from 129 MeteoSwiss stations between 2020 and 2023, I calculate Pearson and Spearman correlations at hourly, daily, and monthly levels. Correlations in the full sample are generally weak, while restricting the analysis to observations above both potential-production thresholds yields negative correlations at each time scale. Subgroup results also suggest variation by season, location, and elevation. These findings indicate that observed complementarity depends on the time scale and sample definition; they describe meteorological associations rather than electricity-production profiles.

Keywords: wind; solar radiation; complementarity; Switzerland.
JEL classification: surely not useful

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Introduction

I explore the geographical and temporal conditions under which wind speed and solar radiation are complementary in Switzerland.

I collect wind and solar radiation data from MeteoSwiss at hourly, daily and monthly levels, including information on season, location and elevation. These variables are of interest because they underlie wind and solar electricity production. Cauz et al. [2025] explore complementarity between electricity production from wind turbines and solar panels (PVs), and find a negative correlation between the two quantities, mitigating the risks to which these renewable energy sources are exposed.

Overall, I find that wind–solar complementarity in Switzerland depends on the time horizon, season, and location. Negative correlation is stronger in the subsample of potential electricity production zones (combination of strong winds and high solar radiation). Finally, wind–solar complementarity is present across the country, but is stronger in the northern part of the country and when elevation reaches 1200 meters.

Across hourly, daily, and monthly observations, Spearman correlations are 0.1333, 0.0770 and 0.0936 in the full sample and -0.0965, -0.1580 and -0.1439 in the threshold-restricted sample. The threshold-restricted estimates are more negative, but they describe associations between meteorological observations—not electricity production or guaranteed supply complementarity.

The paper first describes the data and measures of complementarity (section 1 and section 2), then presents the results (section 3), discusses sensitivity and limitations (subsection 3.1 and section 4), and concludes (section 5).

1 Data and sample construction

I use data from MeteoSwiss from 2020 to 2023[2], relying on measures of mean wind speed (in m/s) and solar radiation (in W/m^2) at hourly, daily and monthly levels across Switzerland. The choice of variables constitutes a hypothesis of this work, which I defend in another document. In short, electricity production from wind turbines depends on wind speed, while solar radiation is the source of electricity production for PVs.

I use data from 129 stations across the country.

Stations are grouped into four elevation classes: <400 m, 400–<800 m, 800–<1,200 m, and ≥1,200 m. GI calculate the median latitude and longitude across the 129 stations in the analysis sample. Each station is assigned to North or South according to whether its latitude is above or below the median, or to East or West according to its longitude; the assignment is based on whichever coordinate differs more from its median. These groups are analytical subdivisions, not official climatic or administrative regions.

Below, I display a map showing station locations across Switzerland, divided into four areas.

[2] I selected this period to align with the work of Cauz et al. [2025], which provides a benchmark.

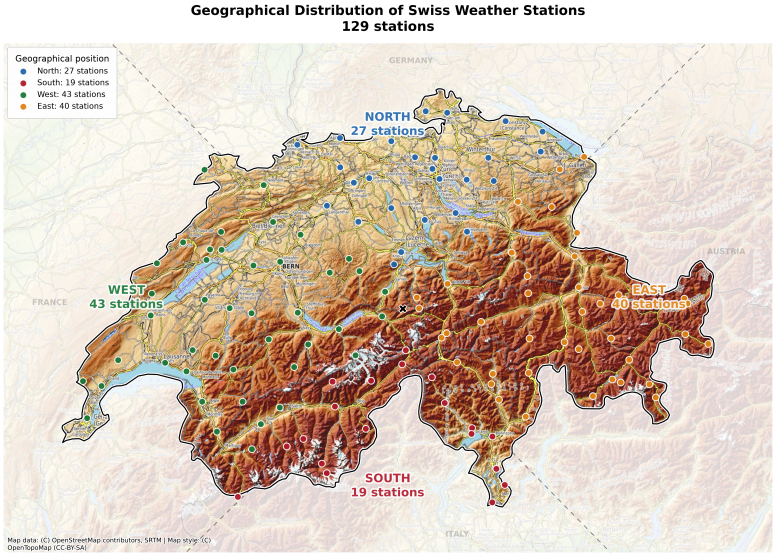

Figure 1: Geographical distribution of Swiss Weather Stations

An important point concerning wind and solar energy is the existence of production thresholds. When wind speed or solar radiation is too low, neither technology can produce electricity. I use 3 m/s for wind speed and 100 W/m^2 for solar radiation as analytical screening thresholds for potential production, rather than as universal technical cut-offs. Actual production depends on the turbine or PV system, so results using these thresholds should be interpreted as conditional on these assumptions. That is why, in the rest of my analysis, I compare, whenever necessary, the whole sample (without thresholds) with the subsample that includes only values above the thresholds (wind speeds higher than 3 m/s AND solar radiation above 100 W/m^2).

On average, across all hours from 2020 to 2023, 29.4% of wind observations have the potential to allow a turbine to start producing electricity, while 61.9% of solar radiation observations have the potential to support electricity production by PVs. Appendix A presents descriptive statistics for both measures across time, geographical zones, and elevation. re compute again the code to check figures!! Add here the number of observationz at hourly, daily and monthly, with or without thresholds.

2 Measuring resource complementarity

Using hourly, daily and monthly data, I compute Spearman and Pearson correlations to explore potential complementarity between the two resources. Pearson correlation measures linear association, whereas Spearman correlation measures monotonic association based on ranks.

We distinguish here between data at hourly, daily and monthly levels. Hourly measures identify very short-term variation, making it possible to study complementarity within a day. Daily data identify variation between days (one day may be cloudy and

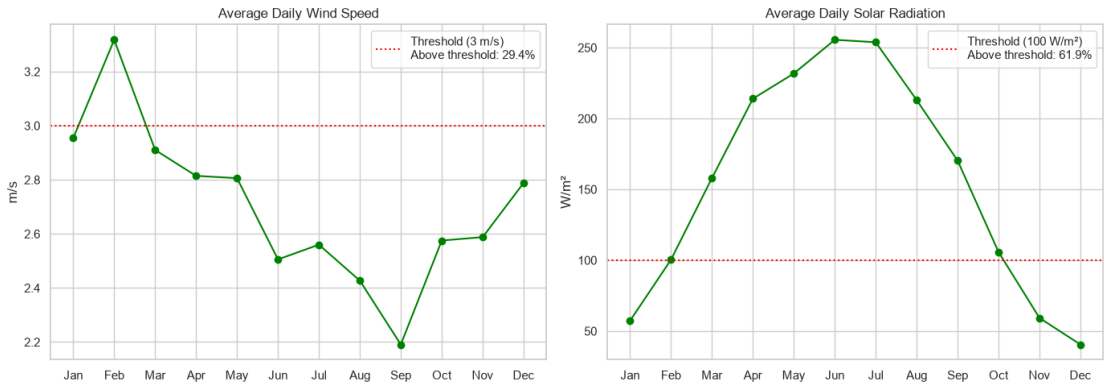

Figure 2: Proportion of measures above the given thresholds

windy, while another is clear with low wind). Monthly data attenuate short-term variation and identify seasonal profiles.

I distinguish seasonal profiles since solar radiation is less likely to be high during winter because of the number of sunny hours and the presence of clouds. I also examine simultaneous observations at the same station and hour using Spearman and Pearson correlations. Finally, I explore the frequency with which a resource exceeds its potential production threshold. For instance, wind could be strong during winter without being strong when solar radiation is low.

I use the entire MeteoSwiss dataset from 2020 to 2023, selecting relevant variables as a baseline, and then apply a filter to create a subsample including only observations above the fixed thresholds (3 m/s for mean wind speed and 100 W/m^2 for mean solar radiation). Thus, I use this subsample for a conditional analysis, in the sense that only these observations allow for potential electricity production. It would make no sense to explore relations between measures when half of the observations are not useful for our purpose – exploring complementarity between wind and solar radiation for wind turbine and PV installations. In doing so, I filter out cases where wind speed is above 3 m/s and solar radiation is below 100 W/m^2. I am not exploring cases where only wind turbines produce electricity (especially at night) or where only PVs produce electricity. That is why I display results with and without thresholds when relevant.

Because this restriction excludes observations in which only one resource meets its threshold, the resulting correlations describe co-occurring above-threshold conditions rather than complementarity across all weather conditions. More negative correlations in this subsample therefore do not, on their own, establish stronger overall complementarity.

A complementary analysis could retain observations for which at least one resource meets its threshold, thereby including cases in which only wind or only solar radiation permits potential electricity production. A future part to include in this work I think!

An important point to note when thresholds are applied is that, at the hourly level, I only keep observations above both thresholds, while at the daily level, I keep an observation only if both daily means are above their respective thresholds. Thus, during the day,

both thresholds are met, but at night, mean solar radiation drops below the threshold, excluding the observation for the whole day. Thus, I keep this day in the hourly analysis, but not in the daily one. The same process is applied to the monthly measurements.

3 Results: overall complementarity

My findings are summarised in Table 1 and Table 2. Additional results are available in subsection A.1, including seasonal, geographical and elevation controls discussed later in this paper.

Table 1: With Threshold

Level	Pearson	Spearman
Hourly	-0.1258	-0.0965
Daily	-0.1719	-0.1580
Monthly	-0.2336	-0.1439

Table 2: Without Threshold

Level	Pearson	Spearman
Hourly	0.0925	0.1333
Daily	-0.0459	0.0770
Monthly	-0.0180	0.0936

Table 3: Cauz et al. (2025)

Level	Spearman
Hourly	-0.176
Daily	-0.368
Monthly	-0.736

The comparison with Cauz et al. is indicative rather than directly like-for-like: their estimates use national electricity-generation profiles, whereas mine use pooled station-level meteorological observations, which differ in unit of observation and normalisation.

Without thresholds, the Pearson correlation is positive at the hourly level and slightly negative at the daily and monthly levels, while the Spearman correlation is always positive but weak. In the filtered subsample, the Pearson and Spearman correlations are negative. For the whole dataset, we remain cautious when it comes to complementarity, since Spearman correlations are not negative at any of these levels.

As in Cauz et al. [2025, Figure 1], I display the different combinations of wind and solar radiation in four zones to illustrate this complementarity in a different way.

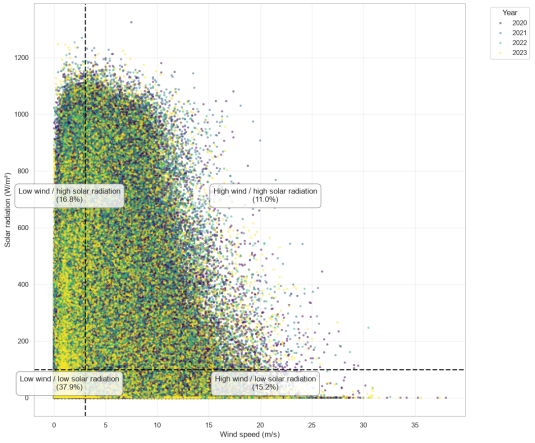

Figure 3: Proportion of wind and radiation conditional on thresholds

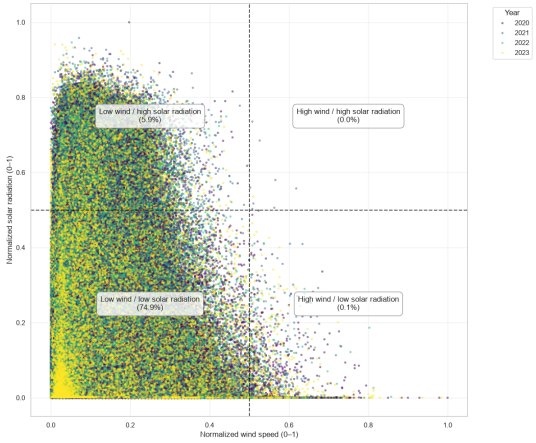

Figure 4: Proportion of wind and radiation without thresholds

Here I need to make a comparison and write a paragraph for some insights

3.1 Results: temporal and seasonal patterns

As noted earlier in Figure 2, both mean wind speed and solar radiation have a seasonal component. In both the whole sample and the subsample, we find that mean wind speed is higher in winter, while mean solar radiation is higher in summer and spring than in other seasons (Table 4 and Figure 5). Thus, the two measures have a strong seasonal component, suggesting complementarity between seasons. Combining wind and solar electricity production is therefore of interest given seasonal meteorological differences. This suggests – as we will see later – a negative correlation between wind and solar measures during cold months (usually December to May, as suggested by a previous analysis of mine). This result is shown in Table 5, where the hourly and daily samples with thresholds display slightly negative correlations in all seasons, especially during autumn and winter.

Table 4: Seasonal wind speed and solar radiation statistics based on daily observations, with and without thresholds.

Sample	Season	Mean wind speed (m/s)	Mean solar radiation (W/m^2)
Without thresholds	Winter	2.90	64.87
Without thresholds	Spring	2.78	201.19
Without thresholds	Summer	2.43	241.03
Without thresholds	Autumn	2.38	111.81
With thresholds	Winter	5.69	132.54
With thresholds	Spring	4.83	226.09
With thresholds	Summer	4.44	257.29
With thresholds	Autumn	4.83	162.41

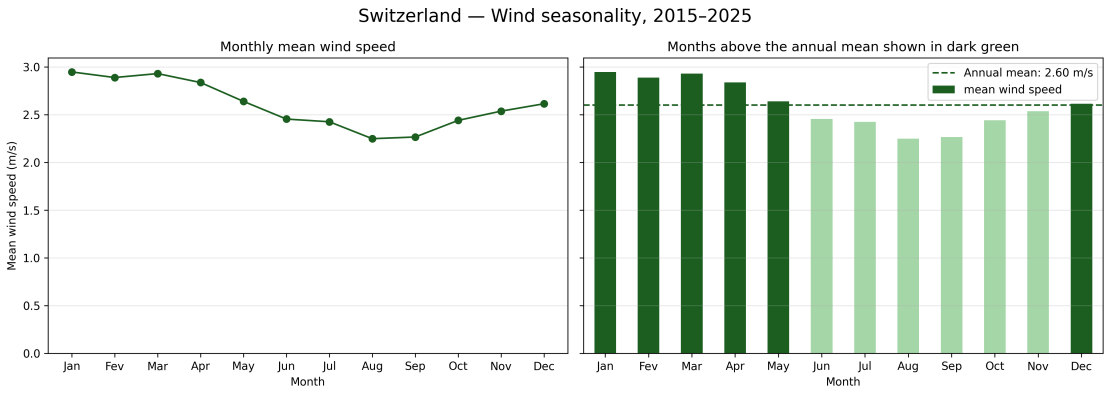

Figure 5: Wind seasonality in Switzerland, 2015–2025

Figure to update at 2020-2023 levels.

Table 5: Global correlations with thresholds

Scale	Season	N	Pearson	Spearman
Hourly	All	604,689	-0.1258	-0.0965
	Winter	88,338	-0.0588	-0.0498
	Spring	212,728	-0.0717	-0.0465
	Summer	199,963	-0.0501	-0.0267
	Autumn	103,660	-0.1436	-0.1191
Daily	All	32,108	-0.1713	-0.1578
	Winter	3,278	-0.0431	-0.0372
	Spring	12,929	-0.1086	-0.0879
	Summer	10,821	-0.1029	-0.0896
	Autumn	5,080	-0.1756	-0.1815
Monthly	All	1,222	-0.2392	-0.1477
	Winter	108	0.0588	0.1435
	Spring	523	0.0050	0.0896
	Summer	382	-0.1083	-0.0861
	Autumn	209	-0.3089	-0.2406

3.2 Results: altitude and geographical heterogeneity

Altitude matters for higher wind speeds, while – except for the presence of clouds – elevation does not matter for solar radiation. Overall, elevations above 1200 meters exhibit a stronger negative correlation between wind and solar radiation. Figure 6 and Figure 7 provide visual evidence of this pattern, with wind speeds increasing on average with elevation.

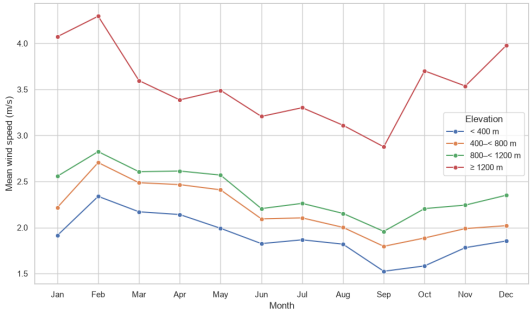

Figure 6: Mean wind speed in whole sample

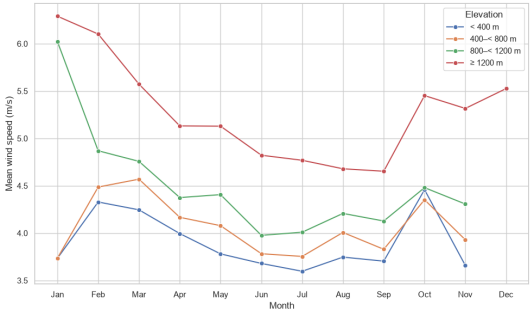

Figure 7: Mean wind speed in the subsample

These stronger winds imply a stronger negative correlation between wind and solar

radiation, as displayed in Table 6. For both Pearson and Spearman correlations, in the whole sample and in the subsample, all correlations above 1200 meters are negative, suggesting wind–solar complementarity.

Table 6: Pearson and Spearman correlations by altitude class

Scale	**Altitude**	**Without Threshold**		**With Threshold**	
		Pearson	Spearman	Pearson	Spearman
Hourly	$<$ 400 m	0.2513	0.2548	-0.0344	-0.0302
	400–$<$ 800 m	0.2016	0.2413	-0.0124	0.0033
	800–$<$ 1200 m	0.1238	0.2118	-0.1497	-0.1193
	$\geq$ 1200 m	-0.0111	-0.0304	-0.1753	-0.1543
Daily	$<$ 400 m	0.0070	0.1187	-0.1458	-0.1356
	400–$<$ 800 m	-0.0206	0.0634	-0.1205	-0.1086
	800–$<$ 1200 m	-0.0421	0.0901	-0.1002	-0.0853
	$\geq$ 1200 m	-0.1323	-0.0174	-0.2009	-0.1939
Monthly	$<$ 400 m	-0.0096	0.0284	-0.1147	-0.1084
	400–$<$ 800 m	0.0253	0.0678	-0.1310	-0.0978
	800–$<$ 1200 m	-0.0538	0.0545	-0.1566	-0.1090
	$\geq$ 1200 m	-0.1248	-0.0082	-0.3190	-0.2412

Elevation depends on a country's topography. This is why I cross topographic data with location (North, South, East, West), providing a broader perspective for the analysis. Results suggest a stronger negative correlation in the northern and western parts of the country. I display correlations using heatmaps in Figure 8 and Figure 9 below. It appears that, in the whole sample, the negative wind–solar correlation is stronger above 1200 meters in the northern part of the country, while the subsample displays stronger negative correlations at elevations from 800 m to 1200 m and above in northern and western areas.

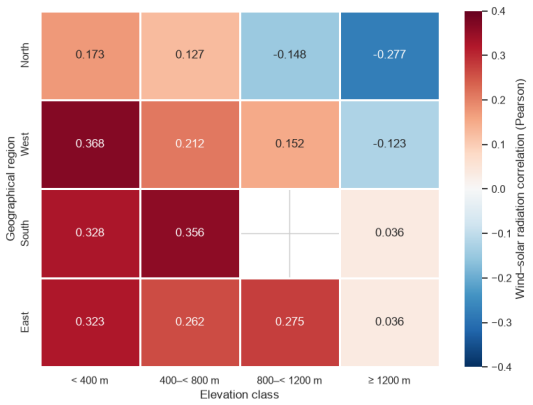

Figure 8: Mean wind speed across months and elevation, without threshold

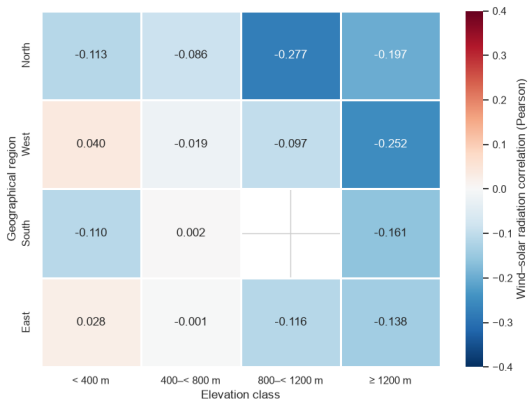

Figure 9: Wind–solar correlation across altitude and location, with threshold

Both existing heatmaps use hourly Pearson correlations in the notebook, not Spearman correlations[3].

4 Sensitivity analysis and limitations

Overall, results are sensitive to thresholds and the method used to compute correlations. Analyses with thresholds yield stronger negative correlations but examine meteorological observations with the potential to support electricity production. Pearson and Spearman correlations distinguish between linear and monotonic associations without establishing complementarity.

Fixed effects regressions distinguish between sources of variation. Station fixed effects (FEs) control for time-invariant mean differences across station locations, while time FEs control for variation common to all stations in each observed period. Thus, estimated coefficients describe a conditional relation, not a causal one. A change in sign after the introduction of FE suggests common variation across stations.

For each temporal resolution, I regress solar radiation on wind speed using a simple specification, a station fixed-effects specification, a time fixed-effects specification, and a specification including both sets of fixed effects. The wind-speed coefficient measures the change in solar radiation associated with a one-m/s increase in wind speed; standard errors are two-way clustered by station and time. Regression results are reported in Table 8 and Table 7.

In addition, correlations within station groups encompass variation that is not visible at the aggregate level. That is why it is important to distinguish stations by altitude and location. Estimates for small subgroups, particularly at the monthly level, should be interpreted cautiously. For example, the threshold-restricted winter group below 400 m contains only two observations, making its correlation estimate uninformative about

[3] I need to add the Spearman's one!

a general relationship.

Additional checks remain necessary concerning variation in thresholds, the final number of observations, especially for monthly measures, and the years studied, *etc.*

We should bear in mind that we are studying complementarity between meteorological resources, not between electricity production profiles. Thus, we need to remain cautious when drawing conclusions about complementarity between wind turbines and PV installations.

5 Conclusion

This analysis shows opposing seasonal profiles for mean wind speed and solar radiation in Switzerland. Wind is, on average, stronger during winter, while solar radiation is low, and vice versa during summer. This suggests potential seasonal complementarity without short-term guarantees.

Results also reveal heterogeneity by elevation and location. Negative correlations at high elevations in the northern and western parts of the country suggest potential complementarity – strongly dependent on topographic location.

In the whole sample, correlations remain weak, while filtered data display stronger negative correlations. Thus, the results do not establish systematically strong and uniform complementarity in Switzerland.

These results favour a precise analysis by location to identify the most favourable combinations. The value of this work lies in a future work: converting meteorological resources into potential electricity production to assess the stability of electricity production and its contribution to demand coverage.

A Descriptive statistics

A.1 Additional controls on correlation measures

Table 7: Wind/Solar regression results by geographical zone

Scale	Model	Zone	Beta	Standard Error	P-value
1. Hourly	Simple	North	7.3522	5.1234	1.5128e-01
		South	14.1971	6.1982	2.1991e-02
		West	6.3674	3.7379	8.8482e-02
		East	10.7395	3.9559	6.6315e-03
	Station FE	North	9.5663	6.1728	1.2120e-01
		South	18.2450	8.1525	2.5225e-02
		West	8.5833	5.1065	9.2789e-02
		East	12.8311	5.3767	1.7013e-02
	Time FE	North	0.0971	0.3355	7.7229e-01
		South	4.8450	0.9729	6.3661e-07
		West	1.4337	0.4152	5.5366e-04
		East	3.4087	0.6029	1.5727e-08
	Station + Time FE	North	0.0671	0.4794	8.8876e-01
		South	5.6508	1.1528	9.4895e-07
		West	1.1053	0.5793	5.6380e-02
		East	2.1561	0.6173	4.7762e-04
2. Daily	Simple	North	-8.9242	1.2305	4.1100e-13
		South	1.9354	2.8243	4.9319e-01
		West	-4.1680	0.8998	3.6215e-06
		East	0.8026	1.4888	5.8983e-01
	Station FE	North	-13.6621	1.8426	1.2257e-13
		South	1.7499	4.6915	7.0915e-01
		West	-9.6085	1.6934	1.3965e-08
		East	-3.7039	2.4412	1.2921e-01
	Time FE	North	0.2595	0.4084	5.2514e-01
		South	4.7154	1.1160	2.3865e-05
		West	1.1119	0.5175	3.1650e-02
		East	4.0724	0.7777	1.6409e-07
	Station + Time FE	North	0.1980	0.6658	7.6614e-01
		South	6.2364	1.4114	9.9406e-06
		West	0.1899	0.5582	7.3366e-01
		East	2.2578	0.7780	3.7102e-03
3. Monthly	Simple	North	-11.0312	3.3376	9.5478e-04
		South	2.8684	4.0465	4.7844e-01
		West	-3.2094	1.7084	6.0347e-02
		East	3.2070	2.2461	1.5339e-01
	Station FE	North	-38.3589	9.8386	9.7715e-05
		South	5.2669	19.6462	7.8864e-01
		West	-24.8229	7.3995	7.9949e-04
		East	-8.6941	9.2917	3.4947e-01
	Time FE	North	-0.4438	0.6862	5.1778e-01
		South	3.1088	1.4876	3.6679e-02
		West	1.5400	0.9187	9.3727e-02
		East	5.3984	1.0682	4.4601e-07
	Station + Time FE	North	-2.0719	2.2962	3.6694e-01
		South	5.8456	2.3956	1.4711e-02
		West	-0.6744	1.1654	5.6282e-01
		East	2.5934	1.2090	3.1985e-02

Table 8: Regression results of the Wind/Solar relationship

Scale	Model	Beta	Standard Error	P-value	Obs (N)
1. Hourly	Simple	9.0325	2.3380	1.1187e-04	4,453,593
	FE Station	11.4943	3.0639	1.7575e-04	4,453,593
	FE Time	2.5051	0.3366	9.8810e-14	4,453,593
	FE Station + Time	1.9891	0.3948	4.7044e-07	4,453,593
2. Daily	Simple	-2.2008	0.7790	4.7261e-03	185,736
	FE Station	-6.8185	1.3843	8.4214e-07	185,736
	FE Time	2.6294	0.4566	8.4636e-09	185,736
	FE Station + Time	1.7456	0.4989	4.6727e-04	185,736
3. Monthly	Simple	-0.9140	1.8107	6.1371e-01	6,114
	FE Station	-18.1413	7.2374	1.2217e-02	6,114
	FE Time	3.0738	0.7121	1.6095e-05	6,114
	FE Station + Time	1.1248	0.9513	2.3711e-01	6,114

Table 9: decriptive statistics: hourly level, with threshold

	fkl010h0	gre000h0
count	604689.0000	604689.0000
mean	5.1873	459.5605
std	2.2029	257.9068
min	3.0000	100.0000
25%	3.6000	231.0000
50%	4.5000	415.0000
75%	6.1000	665.0000
max	30.5000	1324.0000

Table 10: decriptive statistics: hourly level, without threshold

	fkl010h0	gre000h0
count	5256311.0000	4671553.0000
mean	2.6229	155.2391
std	2.4309	247.0414
min	0.0000	0.0000
25%	1.0000	1.0000
50%	1.8000	6.0000
75%	3.5000	228.0000
max	38.0000	1324.0000

Table 11: Wind and solar radiation correlations with thresholds (Daily)

	Scale	Scope	Group	Season	Observations	Pearson	Spearman
0	Daily	Altitude class	400–< 800 m	All periods	7,708	-0.1205	-0.1086
1	Daily	Altitude class	400–< 800 m	Autumn	835	-0.2357	-0.2180
2	Daily	Altitude class	400–< 800 m	Spring	3,859	-0.0703	-0.0581
3	Daily	Altitude class	400–< 800 m	Summer	2,447	-0.0316	-0.0377
4	Daily	Altitude class	400–< 800 m	Winter	567	-0.0314	-0.0276
5	Daily	Altitude class	800–< 1200 m	All periods	3,255	-0.1002	-0.0853
6	Daily	Altitude class	800–< 1200 m	Autumn	502	-0.1021	-0.1163
7	Daily	Altitude class	800–< 1200 m	Spring	1,392	-0.0451	-0.0292
8	Daily	Altitude class	800–< 1200 m	Summer	1,106	0.0070	0.0032
9	Daily	Altitude class	800–< 1200 m	Winter	255	-0.1325	-0.1288
10	Daily	Altitude class	< 400 m	All periods	1,179	-0.1458	-0.1356
11	Daily	Altitude class	< 400 m	Autumn	110	-0.2601	-0.3180
12	Daily	Altitude class	< 400 m	Spring	576	-0.0698	-0.0551
13	Daily	Altitude class	< 400 m	Summer	359	0.0676	0.0277
14	Daily	Altitude class	< 400 m	Winter	134	-0.1480	-0.0658
15	Daily	Altitude class	≥ 1200 m	All periods	19,963	-0.2009	-0.1939
16	Daily	Altitude class	≥ 1200 m	Autumn	3,633	-0.1934	-0.1940
17	Daily	Altitude class	≥ 1200 m	Spring	7,103	-0.1790	-0.1653
18	Daily	Altitude class	≥ 1200 m	Summer	6,910	-0.1271	-0.1239
19	Daily	Altitude class	≥ 1200 m	Winter	2,317	-0.0860	-0.0766
20	Daily	Geographic zone	East	All periods	12,565	-0.2072	-0.1982
21	Daily	Geographic zone	East	Autumn	1,989	-0.1891	-0.2031
22	Daily	Geographic zone	East	Spring	4,770	-0.1249	-0.1165
23	Daily	Geographic zone	East	Summer	4,485	-0.1353	-0.1096
24	Daily	Geographic zone	East	Winter	1,321	-0.0746	-0.0835
25	Daily	Geographic zone	North	All periods	4,768	-0.1737	-0.1623
26	Daily	Geographic zone	North	Autumn	719	-0.1973	-0.2060
27	Daily	Geographic zone	North	Spring	2,154	-0.1305	-0.1023
28	Daily	Geographic zone	North	Summer	1,450	-0.1101	-0.1073
29	Daily	Geographic zone	North	Winter	445	-0.0079	-0.0355
30	Daily	Geographic zone	South	All periods	5,651	-0.1936	-0.1639
31	Daily	Geographic zone	South	Autumn	961	-0.1720	-0.1588
32	Daily	Geographic zone	South	Spring	2,088	-0.1644	-0.1165
33	Daily	Geographic zone	South	Summer	1,917	-0.1085	-0.0820
34	Daily	Geographic zone	South	Winter	685	-0.0673	-0.0278
35	Daily	Geographic zone	West	All periods	9,121	-0.1414	-0.1234
36	Daily	Geographic zone	West	Autumn	1,411	-0.1648	-0.1594
37	Daily	Geographic zone	West	Spring	3,918	-0.0928	-0.0679
38	Daily	Geographic zone	West	Summer	2,970	-0.0883	-0.0869
39	Daily	Geographic zone	West	Winter	822	-0.0177	0.0053
40	Daily	Overall	All stations	All periods	32,105	-0.1719	-0.1580
41	Daily	Overall	All stations	Autumn	5,080	-0.1748	-0.1806
42	Daily	Overall	All stations	Spring	12,930	-0.1103	-0.0887
43	Daily	Overall	All stations	Summer	10,822	-0.1028	-0.0895
44	Daily	Overall	All stations	Winter	3,273	-0.0437	-0.0379

Table 12: Wind and solar radiation correlations with thresholds (Hourly)

	Scale	Scope	Group	Season	Observations	Pearson	Spearman
45	Hourly	Altitude class	400–< 800 m	All periods	205,930	-0.0124	0.0033
46	Hourly	Altitude class	400–< 800 m	Autumn	32,079	-0.0659	-0.0440
47	Hourly	Altitude class	400–< 800 m	Spring	80,265	0.0152	0.0258
48	Hourly	Altitude class	400–< 800 m	Summer	68,652	0.0808	0.0813
49	Hourly	Altitude class	400–< 800 m	Winter	24,934	-0.0264	-0.0306
50	Hourly	Altitude class	800–< 1200 m	All periods	75,620	-0.1497	-0.1193
51	Hourly	Altitude class	800–< 1200 m	Autumn	12,420	-0.1399	-0.1297
52	Hourly	Altitude class	800–< 1200 m	Spring	27,770	-0.0849	-0.0570
53	Hourly	Altitude class	800–< 1200 m	Summer	26,600	-0.0579	-0.0291
54	Hourly	Altitude class	800–< 1200 m	Winter	8,830	-0.0649	-0.0724
55	Hourly	Altitude class	< 400 m	All periods	43,634	-0.0344	-0.0302
56	Hourly	Altitude class	< 400 m	Autumn	6,002	-0.0785	-0.0814
57	Hourly	Altitude class	< 400 m	Spring	17,473	0.0156	0.0230
58	Hourly	Altitude class	< 400 m	Summer	14,348	0.0631	0.0644
59	Hourly	Altitude class	< 400 m	Winter	5,811	0.0401	0.0249
60	Hourly	Altitude class	≥ 1200 m	All periods	279,505	-0.1753	-0.1543
61	Hourly	Altitude class	≥ 1200 m	Autumn	53,159	-0.1867	-0.1683
62	Hourly	Altitude class	≥ 1200 m	Spring	87,220	-0.1372	-0.1168
63	Hourly	Altitude class	≥ 1200 m	Summer	90,363	-0.1000	-0.0823
64	Hourly	Altitude class	≥ 1200 m	Winter	48,763	-0.1156	-0.1073
65	Hourly	Geographic zone	East	All periods	210,337	-0.1319	-0.1054
66	Hourly	Geographic zone	East	Autumn	36,282	-0.1554	-0.1319
67	Hourly	Geographic zone	East	Spring	71,700	-0.0766	-0.0633
68	Hourly	Geographic zone	East	Summer	71,494	-0.0639	-0.0448
69	Hourly	Geographic zone	East	Winter	30,861	-0.0621	-0.0508
70	Hourly	Geographic zone	North	All periods	101,983	-0.1505	-0.1320
71	Hourly	Geographic zone	North	Autumn	16,874	-0.1392	-0.1163
72	Hourly	Geographic zone	North	Spring	38,268	-0.0925	-0.0674
73	Hourly	Geographic zone	North	Summer	30,030	-0.0583	-0.0498
74	Hourly	Geographic zone	North	Winter	16,811	-0.0678	-0.0549
75	Hourly	Geographic zone	South	All periods	93,126	-0.1386	-0.1136
76	Hourly	Geographic zone	South	Autumn	15,789	-0.1722	-0.1486
77	Hourly	Geographic zone	South	Spring	31,665	-0.0889	-0.0713
78	Hourly	Geographic zone	South	Summer	31,960	-0.0594	-0.0505
79	Hourly	Geographic zone	South	Winter	13,712	-0.0867	-0.0787
80	Hourly	Geographic zone	West	All periods	199,243	-0.1389	-0.0953
81	Hourly	Geographic zone	West	Autumn	34,715	-0.1576	-0.1281
82	Hourly	Geographic zone	West	Spring	71,095	-0.0803	-0.0365
83	Hourly	Geographic zone	West	Summer	66,479	-0.0600	-0.0155
84	Hourly	Geographic zone	West	Winter	26,954	-0.0888	-0.0768
85	Hourly	Overall	All stations	All periods	604,689	-0.1258	-0.0965
86	Hourly	Overall	All stations	Autumn	103,660	-0.1436	-0.1191
87	Hourly	Overall	All stations	Spring	212,728	-0.0717	-0.0465
88	Hourly	Overall	All stations	Summer	199,963	-0.0501	-0.0267
89	Hourly	Overall	All stations	Winter	88,338	-0.0588	-0.0498

Table 13: Wind and solar radiation correlations with thresholds (Monthly)

	Scale	Scope	Group	Season	Observations	Pearson	Spearman
90	Monthly	Altitude class	400–< 800 m	All periods	222	-0.1310	-0.0978
91	Monthly	Altitude class	400–< 800 m	Autumn	15	-0.2506	-0.2286
92	Monthly	Altitude class	400–< 800 m	Spring	151	-0.1302	-0.0924
93	Monthly	Altitude class	400–< 800 m	Summer	48	-0.0334	-0.0098
94	Monthly	Altitude class	400–< 800 m	Winter	8	0.0286	0.1190
95	Monthly	Altitude class	800–< 1200 m	All periods	109	-0.1566	-0.1090
96	Monthly	Altitude class	800–< 1200 m	Autumn	14	-0.4384	-0.4725
97	Monthly	Altitude class	800–< 1200 m	Spring	60	-0.2099	-0.1789
98	Monthly	Altitude class	800–< 1200 m	Summer	27	0.1231	0.0714
99	Monthly	Altitude class	800–< 1200 m	Winter	8	-0.2253	-0.3333
100	Monthly	Altitude class	< 400 m	All periods	18	-0.1147	-0.1084
101	Monthly	Altitude class	< 400 m	Spring	8	-0.0084	-0.0238
102	Monthly	Altitude class	< 400 m	Summer	8	0.5205	0.6667
103	Monthly	Altitude class	< 400 m	Winter	2	1.0000	1.0000
104	Monthly	Altitude class	≥ 1200 m	All periods	871	-0.3190	-0.2412
105	Monthly	Altitude class	≥ 1200 m	Autumn	180	-0.3641	-0.3049
106	Monthly	Altitude class	≥ 1200 m	Spring	304	-0.2298	-0.1239
107	Monthly	Altitude class	≥ 1200 m	Summer	299	-0.1208	-0.1029
108	Monthly	Altitude class	≥ 1200 m	Winter	88	-0.2193	-0.1737
109	Monthly	Geographic zone	East	All periods	507	-0.2742	-0.1580
110	Monthly	Geographic zone	East	Autumn	93	-0.3741	-0.2548
111	Monthly	Geographic zone	East	Spring	186	0.0640	0.1642
112	Monthly	Geographic zone	East	Summer	176	-0.1811	-0.1328
113	Monthly	Geographic zone	East	Winter	52	0.2718	0.2645
114	Monthly	Geographic zone	North	All periods	174	-0.3506	-0.3200
115	Monthly	Geographic zone	North	Autumn	30	-0.5946	-0.5519
116	Monthly	Geographic zone	North	Spring	83	-0.2523	-0.2568
117	Monthly	Geographic zone	North	Summer	57	0.0926	0.0432
118	Monthly	Geographic zone	North	Winter	4	0.1416	0.2000
119	Monthly	Geographic zone	South	All periods	233	-0.2830	-0.2061
120	Monthly	Geographic zone	South	Autumn	43	-0.2217	-0.1924
121	Monthly	Geographic zone	South	Spring	87	0.0082	0.0406
122	Monthly	Geographic zone	South	Summer	78	-0.1789	-0.1781
123	Monthly	Geographic zone	South	Winter	25	-0.1811	-0.1808
124	Monthly	Geographic zone	West	All periods	306	-0.1907	-0.1069
125	Monthly	Geographic zone	West	Autumn	43	-0.3438	-0.3143
126	Monthly	Geographic zone	West	Spring	167	-0.0438	0.0226
127	Monthly	Geographic zone	West	Summer	71	-0.2301	-0.2170
128	Monthly	Geographic zone	West	Winter	25	0.1205	0.2754
129	Monthly	Overall	All stations	All periods	1,220	-0.2336	-0.1439
130	Monthly	Overall	All stations	Autumn	209	-0.3102	-0.2420
131	Monthly	Overall	All stations	Spring	523	0.0054	0.0903
132	Monthly	Overall	All stations	Summer	382	-0.1098	-0.0855
133	Monthly	Overall	All stations	Winter	106	0.0946	0.1785

Table 14: Wind and solar radiation correlations without thresholds (Daily)

	Scale	Scope	Group	Season	Observations	Pearson	Spearman
0	Daily	Altitude class	400–< 800 m	All periods	95,339	-0.0206	0.0634
1	Daily	Altitude class	400–< 800 m	Autumn	23,749	-0.1269	-0.0822
2	Daily	Altitude class	400–< 800 m	Spring	24,000	-0.0529	-0.0171
3	Daily	Altitude class	400–< 800 m	Summer	24,011	0.0825	0.0889
4	Daily	Altitude class	400–< 800 m	Winter	23,579	-0.0683	-0.0637
5	Daily	Altitude class	800–< 1200 m	All periods	36,506	-0.0421	0.0901
6	Daily	Altitude class	800–< 1200 m	Autumn	9,084	-0.1233	-0.0314
7	Daily	Altitude class	800–< 1200 m	Spring	9,197	-0.0246	0.0747
8	Daily	Altitude class	800–< 1200 m	Summer	9,200	0.1515	0.2432
9	Daily	Altitude class	800–< 1200 m	Winter	9,025	-0.1232	-0.1018
10	Daily	Altitude class	< 400 m	All periods	24,837	0.0070	0.1187
11	Daily	Altitude class	< 400 m	Autumn	6,188	-0.1020	-0.0336
12	Daily	Altitude class	< 400 m	Spring	6,256	0.0010	0.0530
13	Daily	Altitude class	< 400 m	Summer	6,256	0.0175	0.0273
14	Daily	Altitude class	< 400 m	Winter	6,137	-0.0018	0.0342
15	Daily	Altitude class	≥ 1200 m	All periods	72,972	-0.1323	-0.0174
16	Daily	Altitude class	≥ 1200 m	Autumn	18,145	-0.1784	-0.0788
17	Daily	Altitude class	≥ 1200 m	Spring	18,397	-0.0830	0.0285
18	Daily	Altitude class	≥ 1200 m	Summer	18,382	-0.0444	0.0312
19	Daily	Altitude class	≥ 1200 m	Winter	18,048	-0.0474	0.0190
20	Daily	Geographic zone	East	All periods	67,153	0.0187	0.1780
21	Daily	Geographic zone	East	Autumn	16,721	-0.0361	0.0922
22	Daily	Geographic zone	East	Spring	16,916	0.0805	0.1761
23	Daily	Geographic zone	East	Summer	16,910	0.0776	0.1730
24	Daily	Geographic zone	East	Winter	16,606	0.1131	0.1740
25	Daily	Geographic zone	North	All periods	54,440	-0.1458	-0.0702
26	Daily	Geographic zone	North	Autumn	13,559	-0.1703	-0.1507
27	Daily	Geographic zone	North	Spring	13,705	-0.1458	-0.1099
28	Daily	Geographic zone	North	Summer	13,707	-0.0728	-0.0292
29	Daily	Geographic zone	North	Winter	13,469	-0.0491	-0.0393
30	Daily	Geographic zone	South	All periods	32,137	0.0055	0.1758
31	Daily	Geographic zone	South	Autumn	8,006	-0.0612	0.0811
32	Daily	Geographic zone	South	Spring	8,093	0.0746	0.1905
33	Daily	Geographic zone	South	Summer	8,096	0.0195	0.0955
34	Daily	Geographic zone	South	Winter	7,942	0.0813	0.1840
35	Daily	Geographic zone	West	All periods	75,924	-0.0867	0.0239
36	Daily	Geographic zone	West	Autumn	18,880	-0.1419	-0.0846
37	Daily	Geographic zone	West	Spring	19,136	-0.0880	-0.0181
38	Daily	Geographic zone	West	Summer	19,136	0.0017	0.0839
39	Daily	Geographic zone	West	Winter	18,772	-0.0505	-0.0586
40	Daily	Overall	All stations	All periods	229,654	-0.0461	0.0770
41	Daily	Overall	All stations	Autumn	57,166	-0.0936	-0.0136
42	Daily	Overall	All stations	Spring	57,850	-0.0170	0.0574
43	Daily	Overall	All stations	Summer	57,849	0.0150	0.0851
44	Daily	Overall	All stations	Winter	56,789	0.0293	0.0536

Table 15: Wind and solar radiation correlations without thresholds (Hourly)

	Scale	Scope	Group	Season	Observations	Pearson	Spearman
45	Hourly	Altitude class	400–< 800 m	All periods	2,288,018	0.2016	0.2413
46	Hourly	Altitude class	400–< 800 m	Autumn	569,966	0.1443	0.1908
47	Hourly	Altitude class	400–< 800 m	Spring	575,927	0.2522	0.3072
48	Hourly	Altitude class	400–< 800 m	Summer	576,239	0.3423	0.3778
49	Hourly	Altitude class	400–< 800 m	Winter	565,886	0.0553	0.0701
50	Hourly	Altitude class	800–< 1200 m	All periods	876,079	0.1238	0.2118
51	Hourly	Altitude class	800–< 1200 m	Autumn	217,982	0.0793	0.1650
52	Hourly	Altitude class	800–< 1200 m	Spring	220,697	0.1805	0.2786
53	Hourly	Altitude class	800–< 1200 m	Summer	220,800	0.2780	0.3493
54	Hourly	Altitude class	800–< 1200 m	Winter	216,600	-0.0114	0.0450
55	Hourly	Altitude class	< 400 m	All periods	596,088	0.2513	0.2548
56	Hourly	Altitude class	< 400 m	Autumn	148,512	0.1787	0.1682
57	Hourly	Altitude class	< 400 m	Spring	150,144	0.3262	0.3466
58	Hourly	Altitude class	< 400 m	Summer	150,144	0.3741	0.3888
59	Hourly	Altitude class	< 400 m	Winter	147,288	0.1095	0.0852
60	Hourly	Altitude class	≥ 1200 m	All periods	1,751,100	-0.0111	-0.0304
61	Hourly	Altitude class	≥ 1200 m	Autumn	435,347	-0.0259	-0.0704
62	Hourly	Altitude class	≥ 1200 m	Spring	441,502	0.0186	0.0236
63	Hourly	Altitude class	≥ 1200 m	Summer	441,121	0.0699	0.0836
64	Hourly	Altitude class	≥ 1200 m	Winter	433,130	-0.0231	-0.1255
65	Hourly	Geographic zone	East	All periods	1,611,502	0.1183	0.1412
66	Hourly	Geographic zone	East	Autumn	401,254	0.0728	0.0737
67	Hourly	Geographic zone	East	Spring	405,912	0.1686	0.2177
68	Hourly	Geographic zone	East	Summer	405,793	0.2207	0.2671
69	Hourly	Geographic zone	East	Winter	398,543	0.0504	-0.0224
70	Hourly	Geographic zone	North	All periods	1,306,472	0.0472	0.1079
71	Hourly	Geographic zone	North	Autumn	325,416	0.0249	0.0644
72	Hourly	Geographic zone	North	Spring	328,888	0.1026	0.1679
73	Hourly	Geographic zone	North	Summer	328,943	0.1590	0.2346
74	Hourly	Geographic zone	North	Winter	323,225	0.0199	0.0055
75	Hourly	Geographic zone	South	All periods	771,213	0.0949	0.0785
76	Hourly	Geographic zone	South	Autumn	192,095	0.0514	0.0209
77	Hourly	Geographic zone	South	Spring	194,206	0.1446	0.1498
78	Hourly	Geographic zone	South	Summer	194,304	0.1762	0.1799
79	Hourly	Geographic zone	South	Winter	190,608	0.0273	-0.0711
80	Hourly	Geographic zone	West	All periods	1,822,098	0.0879	0.1688
81	Hourly	Geographic zone	West	Autumn	453,042	0.0547	0.1346
82	Hourly	Geographic zone	West	Spring	459,264	0.1364	0.2255
83	Hourly	Geographic zone	West	Summer	459,264	0.2134	0.3109
84	Hourly	Geographic zone	West	Winter	450,528	0.0110	0.0130
85	Hourly	Overall	All stations	All periods	5,511,285	0.0925	0.1333
86	Hourly	Overall	All stations	Autumn	1,371,807	0.0564	0.0828
87	Hourly	Overall	All stations	Spring	1,388,270	0.1429	0.1999
88	Hourly	Overall	All stations	Summer	1,388,304	0.1970	0.2580
89	Hourly	Overall	All stations	Winter	1,362,904	0.0304	-0.0129

Table 16: Wind and solar radiation correlations without thresholds (Monthly)

	Scale	Scope	Group	Season	Observations	Pearson	Spearman
90	Monthly	Altitude class	400–< 800 m	All periods	3,133	0.0253	0.0678
91	Monthly	Altitude class	400–< 800 m	Autumn	783	-0.1385	-0.1090
92	Monthly	Altitude class	400–< 800 m	Spring	783	-0.0249	-0.0202
93	Monthly	Altitude class	400–< 800 m	Summer	783	0.1675	0.1748
94	Monthly	Altitude class	400–< 800 m	Winter	784	0.2597	0.2402
95	Monthly	Altitude class	800–< 1200 m	All periods	1,200	-0.0538	0.0545
96	Monthly	Altitude class	800–< 1200 m	Autumn	300	-0.1466	-0.0757
97	Monthly	Altitude class	800–< 1200 m	Spring	300	-0.0786	-0.0230
98	Monthly	Altitude class	800–< 1200 m	Summer	300	0.2496	0.2690
99	Monthly	Altitude class	800–< 1200 m	Winter	300	0.0339	0.0298
100	Monthly	Altitude class	< 400 m	All periods	816	-0.0096	0.0284
101	Monthly	Altitude class	< 400 m	Autumn	204	-0.2192	-0.1653
102	Monthly	Altitude class	< 400 m	Spring	204	-0.2298	-0.2428
103	Monthly	Altitude class	< 400 m	Summer	204	-0.0840	-0.0396
104	Monthly	Altitude class	< 400 m	Winter	204	0.1036	0.0811
105	Monthly	Altitude class	≥ 1200 m	All periods	2,400	-0.1248	-0.0082
106	Monthly	Altitude class	≥ 1200 m	Autumn	600	-0.1392	-0.0276
107	Monthly	Altitude class	≥ 1200 m	Spring	600	0.0426	0.1400
108	Monthly	Altitude class	≥ 1200 m	Summer	600	-0.0154	0.0235
109	Monthly	Altitude class	≥ 1200 m	Winter	600	0.1141	0.1762
110	Monthly	Geographic zone	East	All periods	2,208	0.0719	0.2191
111	Monthly	Geographic zone	East	Autumn	552	0.0631	0.1578
112	Monthly	Geographic zone	East	Spring	552	0.3330	0.3609
113	Monthly	Geographic zone	East	Summer	552	0.1908	0.2449
114	Monthly	Geographic zone	East	Winter	552	0.3199	0.3703
115	Monthly	Geographic zone	North	All periods	1,789	-0.1573	-0.0905
116	Monthly	Geographic zone	North	Autumn	447	-0.1647	-0.1688
117	Monthly	Geographic zone	North	Spring	447	-0.1819	-0.1892
118	Monthly	Geographic zone	North	Summer	447	-0.0434	0.0334
119	Monthly	Geographic zone	North	Winter	448	0.2331	0.2645
120	Monthly	Geographic zone	South	All periods	1,056	0.0290	0.1685
121	Monthly	Geographic zone	South	Autumn	264	-0.0229	0.0939
122	Monthly	Geographic zone	South	Spring	264	0.2211	0.2401
123	Monthly	Geographic zone	South	Summer	264	0.0301	-0.0119
124	Monthly	Geographic zone	South	Winter	264	0.2106	0.2531
125	Monthly	Geographic zone	West	All periods	2,496	-0.0668	0.0253
126	Monthly	Geographic zone	West	Autumn	624	-0.0997	-0.0754
127	Monthly	Geographic zone	West	Spring	624	-0.0461	-0.0211
128	Monthly	Geographic zone	West	Summer	624	-0.0200	0.1017
129	Monthly	Geographic zone	West	Winter	624	0.1938	0.2280
130	Monthly	Overall	All stations	All periods	7,549	-0.0184	0.0936
131	Monthly	Overall	All stations	Autumn	1,887	-0.0394	0.0189
132	Monthly	Overall	All stations	Spring	1,887	0.1041	0.1215
133	Monthly	Overall	All stations	Summer	1,887	0.0409	0.0962
134	Monthly	Overall	All stations	Winter	1,888	0.2493	0.2664

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